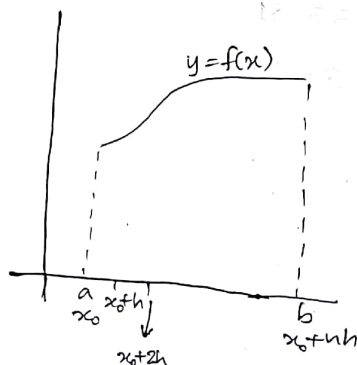


$x_0 - 3h$	y_{-3}
$x_0 - 2h$	y_{-2}
$x_0 - h$	y_{-1}
x_0	y_0
x_1	y_1
x_2	y_2
x_3	y_3

Numerical Integration

Suppose there are n data points $y_0, y_1, y_2, \dots, y_n$ are given for an equally spaced interval $[x_0, x_n]$ (i.e., $x_0, x_0+h, x_0+2h, \dots, x_0+nh$). This integration can be defined as $\int_a^b f(x) dx = \int_{x_0}^{x_0+nh} f(x) dx$.



We can evaluate this integral using Newton's Forward Difference Interpolation Formula. This is known as Newton-Cotes Integration formula.

$$I = \int_{x_0}^{x_0+nh} f(x) dx$$

$$= \int_0^n f(x_0+uh) h du$$

$$= h \int_0^n f(x_0+uh) du$$

$$= h \int_0^n \left\{ y_0 + \frac{u}{1!} \Delta y_0 + \frac{u(u-1)}{2!} \Delta^2 y_0 + \dots \right\} du$$

Integrating on RHS by term by term

$$\therefore I = h \left\{ y_0 \int_0^n du + \Delta y_0 \int_0^n u du + \frac{\Delta^2 y_0}{2!} \int_0^n u(u-1) du + \dots \right\}$$

$$= h \left\{ n y_0 + \frac{n^2}{2} \Delta y_0 + \left(\frac{n^3}{3} - \frac{n^2}{2} \right) \frac{\Delta^2 y_0}{2!} + \dots \right\}$$

$$= nh \left\{ y_0 + \frac{n}{2} \Delta y_0 + \frac{2n^2-3n}{12} \Delta^2 y_0 + \frac{n(n-1)^2}{24} \Delta^3 y_0 + \dots \right\} \quad \text{--- (1)}$$

Case 1: (Trapezoidal Rule)

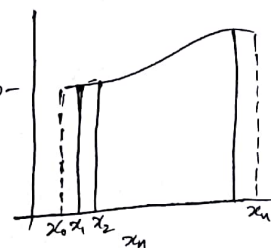
Put $n=1$ in (1)

Here there are only two paired values in each subinterval, hence interpolating polynomial is linear.

$\therefore$ (1) can be written as

$$I = \int_{x_0}^{x_0+nh} f(x) dx = \int_{x_0}^{x_1} f(x) dx + \int_{x_1}^{x_2} f(x) dx + \dots + \int_{x_{n-1}}^{x_n} f(x) dx$$

$$= h \left[y_0 + \frac{\Delta y_0}{2} + y_1 + \frac{\Delta y_1}{2} + \dots + y_{n-1} + \frac{\Delta y_{n-1}}{2} \right]$$



$$\begin{aligned}
 J &= h \left[y_0 + \frac{y_1 - y_0}{2} + y_1 + \frac{y_2 - y_1}{2} + \dots + y_{n-1} + \frac{y_n - y_{n-1}}{2} \right] \\
 &= h \left[\frac{y_0 + y_1}{2} + \frac{y_1 + y_2}{2} + \dots + \frac{y_{n-1} + y_n}{2} \right] \\
 &= \frac{h}{2} \left[(y_0 + y_n) + 2(y_1 + y_2 + \dots + y_{n-1}) \right] \\
 &= \frac{h}{2} \left[(1^{\text{st}} + \text{last}) + 2 \times \text{Sum of the remaining terms} \right]
 \end{aligned}$$

Case 2 Simpson's $\frac{1}{3}$ rd Rule

Put $n=2$ in ①

$$J = \int_{x_0}^{x_0+h} f(x) dx = \frac{h}{3} \left[(y_0 + y_n) + 2(y_2 + y_4 + \dots) + 4(y_1 + y_3 + \dots) \right]$$

Case 3 Simpson's Three-Eight Rule

Put $n=3$ in ①

$$J = \int_{x_0}^{x_0+h} f(x) dx = \frac{3h}{8} \left[(y_0 + y_n) + 3(y_1 + y_2 + y_4 + \dots) + 2(y_3 + y_5 + \dots) \right]$$

Find the approximate value of $\int_0^{\pi} \sin x \, dx$ using trapezoidal rule, Simpson's $\frac{1}{3}$ rd rule, Simpson's $\frac{3}{8}$ rule by dividing the range of integration into 6 equal parts.

x :	0	$\pi/6$	$\pi/3$	$\pi/2$	$2\pi/3$	$5\pi/6$	π
$y = \sin x$:	0.0	0.5	0.8660	1.0	0.8660	0.5	0.0
y_0	y_1	y_2	y_3	y_4	y_5	y_6	

Here $h = \text{width of interval} = \pi/6$

Case 1

$$\begin{aligned}
 J &= \frac{h}{2} \left[(y_0 + y_6) + 2(y_1 + y_2 + y_3 + y_4 + y_5) \right] \\
 &= \frac{\pi/6}{2} \left[(0 + 0) + 2(0.5 + 0.8660 + 1.0 + 0.8660 + 0.5) \right] \\
 &= \frac{\pi}{12} \times 7.464 \\
 &= 0.622\pi
 \end{aligned}$$

$$\begin{aligned}
 &= 1.953 \\
 &= 1.953
 \end{aligned}$$

Case 2

$$\begin{aligned}
 J &= \frac{h}{3} \left[(y_0 + y_6) + 2(y_2 + y_4) + 4(y_1 + y_3 + y_5) \right] \\
 &= \frac{\pi/6}{3} \left[(0 + 0) + 2(0.8660 + 0.8660) + 4(0.5 + 1.0 + 0.5) \right] \\
 &= \frac{\pi}{18} \times 9.732 \\
 &= 1.9998
 \end{aligned}$$

Case 3

$$J = \frac{5h}{8} [(y_0 + y_6) + 3(y_1 + y_2 + y_4 + y_5) + 2(y_3)]$$

$$= \frac{3 \times \pi \times 8}{8} [(0+0) + 3(0.5 + 0.8660 + 0.8660 + 0.5) + 2 \times 1]$$

$$= \frac{\pi}{16} \times 10.196$$

$$= 2.00097$$

Simpson's 1/3rd Rule

Simpson's rule is used when number of parts are even, i.e., number of ordinates are odd.

In trapezoidal rule ~~is~~ $f(x)$ is a

linear function of x .

In Simpson's rule $f(x)$ is a quadratic function of x .

Note

(i) Area bounded by a curve $y=f(x)$ between $x=a$ and $x=b$ and the x -axis is given by

$$\int_a^b y dx = \int_a^b f(x) dx$$

(ii) Volume of solid formed by revolving the curve $y=f(x)$ between $x=a$ and $x=b$ is given

by $V = \int_a^b \pi y^2 dx$

(iii) Length of arc of the curve $y=f(x)$ between $x=a$ and $x=b$ and x -axis is given by $\int_a^b \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx$

Evaluate $\int_{-2}^2 x^3 dx$ using trapezoidal rule by dividing the range into 4 equal parts.

$$a = -2$$

$$b = 2$$

$$h = \frac{b-a}{n}$$

$$= \frac{2 - (-2)}{4}$$

$$= 1$$

$$y(x) = y = x^3$$

x_0	x_1	x_2	x_3	x_4
-2	-1	0	1	2
y_0	y_1	y_2	y_3	y_4
-8	-1	0	1	8

The ~~vel~~ velocity, V (km/min) of a moped which starts from rest is moped which starts from rest is given at fixed intervals of time as follows. Estimate approximately the distance covered in 20 minutes.

x_0	x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8	x_9	x_{10}
0	2	4	6	8	10	12	14	16	18	20
y_0	y_1	y_2	y_3	y_4	y_5	y_6	y_7	y_8	y_9	y_{10}
0	10	18	25	29	32	20	11	5	2	0

We have

$$\frac{ds}{dt} = V$$

$$ds = V dt$$

$$\int_0^{20} ds = \int_0^{20} V dt$$

$$\therefore (s)_0^{20} = \int_0^{20} V dt$$

(Apply Simpson's rule to evaluate the integral)

$$= \frac{h}{3} \left[(y_0 + y_{10}) + 2(y_1 + y_2 + \dots + y_9) + 4(y_3 + y_5 + y_7 + y_9) \right]$$

$$= \frac{10 \times 2}{3} \left[(0+0) + 2(18+29+20+5) + 4(10+25+32+11+2) \right]$$

$$= \frac{2}{3} \times (144 + 320)$$

$$\approx 309.33 \text{ km}$$

The velocity V of a particle at a distance 5 from a point on its path is given by the table.

x_0	x_1	x_2	x_3	x_4	x_5	x_6
0	10	20	30	40	50	60
V	y_1	y_2	y_3	y_4	y_5	y_6
47	58	64	65	61	52	38

Estimate the time taken to cover 60 ft using Simpson's rule.

$$\frac{ds}{dt} = V$$

$$\frac{ds}{V} = dt$$

$$\int_0^{60} dt = \int_0^{60} \frac{ds}{V}$$

$$(t)_0^{60} = \int_0^{60} \frac{ds}{V}$$

$$= \frac{h}{3} \left[\left(\frac{t_0}{V_0} + \frac{t_6}{V_6} \right) + 2 \left(\frac{t_1}{V_1} + \frac{t_2}{V_2} + \frac{t_3}{V_3} + \frac{t_4}{V_4} + \frac{t_5}{V_5} \right) + 4 \left(\frac{t_2}{V_2} + \frac{t_4}{V_4} \right) \right]$$

$$= \frac{10}{3} \left[\left(\frac{47}{38} + \frac{1}{3} \right) + 2 \left(\frac{58}{61} + \frac{64}{52} + \frac{65}{38} \right) + 4 \left(\frac{64}{52} + \frac{65}{38} \right) \right]$$

$$= \frac{10}{3} \left[85.7556 + 1.667 \right]$$

$$= \frac{10}{3} \times 87.4226$$

$$\approx 291.41$$

(i) In Simpson's $1/3$ rule $y(x)$ is a polynomial of degree 2, hence the number of intervals must be even, ie the number of ordinates must be odd.

(ii) In Simpson's $3/8$ rule $y(x)$ is a polynomial of degree 3, hence the number of intervals is a multiple of 3.

Numerical Solution of Differential Equations

(i) Taylor series Method

Taylor series method can be used to solve a differential equation of order one with some initial conditions are given. ie, equations of the form $(dy/dx) = f(x, y)$ with initial condition $y(x_0) = y_0$.

Using Taylor Series for a function $y = f(x)$, find $y(0.1)$ corrected to four decimal places if $y(x)$ satisfies $(dy/dx) = y' = x - y^2$ and $y(0) = 1$

x	0	0.1
$y(x)$	1	

$y_0 = y(x_0)$

$$h = \frac{x_1 - x_0}{n}$$

$$= \frac{0.1 - 0}{1}$$

$$= 0.1$$

From the table it is clear that we have to calculate y_1 .

We have the Taylor series expansion for $y(x)$ as $y(x) = y_0 + (x - x_0)y_0' + \frac{(x - x_0)^2}{2!}y_0'' + \dots$

We put $x = x_1$

$$y(x_1) = y_1 = y_0 + (x_1 - x_0)y_0' + \frac{(x_1 - x_0)^2}{2!}y_0'' + \dots$$

$$y_1 = y_0 + h y_0' + \frac{h^2}{2!} y_0'' + \dots$$

Find y_0, y_0', y_0''

$$y_0 = 1$$

$$y_0' = -1$$

$$y_0'' = 3$$

$$y''(x_0) = y_0'' = 1 - 2 \times 1 \times 1 = -1$$

$$y'(x_0) = y_0' = x_0 - y_0^2$$

$$= 0 - 1 = -1$$

$$y'(x) = x - y^2 = \frac{dy}{dx}$$

$$y''(x) = 1 - 2y \frac{dy}{dx}$$

$$= 1 - 2 \times 1 \times (-1) = 3$$

$$y_0''' = -8$$

We have $y''(x) = 1 - 2y'$

$$y_0'' = 34$$

$$y'''(x) = 0 - 2[y' + y'']$$

$$x = x_0 \quad y'''(x_0) = -2[y_0' + y_0'']$$

$$= -2[34 + 34] = -8$$

$$y_0''' = 34$$

$$y''(x) = -2[y' + y'']$$

$$x = x_0 \quad y''(x_0) = -2[y_0' + y_0'']$$

$$= -2[34 + 34] = -8$$

$$= -2(-8 - 3 - 6) = 34$$

$$y(0.1) = y_1 = y_0 + h y_0' + \frac{h^2}{2!} y_0'' + \frac{h^3}{3!} y_0''' + \frac{h^4}{4!} y_0^{(4)}$$

$$= 1 + (0.1) \times 1 + \frac{(0.1)^2}{2} \times 3 + \frac{(0.1)^3}{6} \times -8 + \frac{(0.1)^4}{24} \times 34$$

$$= 1 + 0.1 + 0.015 + -0.00133 + 0.001416 \times 10^3$$

$$= 0.913812$$

(ii) Euler's Method

Consider a curve as shown in the figure. In a small interval the line PQ is approximating the curve

because x_0, y_0 and $\frac{dy}{dx} = f(x, y)$

is given we can follow the procedure by writing the equation of

the tangent at PQ

$$y_1 - y_0 = \text{slope } (x_1 - x_0)$$

$$= \frac{dy}{dx} (x_1 - x_0)$$

$$= h f(x_0, y_0)$$

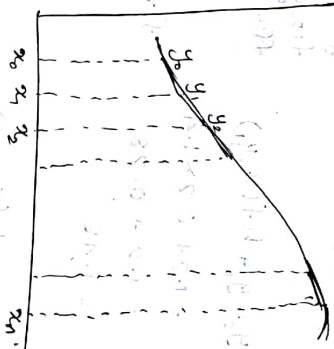
$$y_1 = y_0 + h f(x_0, y_0)$$

h = $x_1 - x_0$
interval length

Similarly

$$y_2 = y_1 + h f(x_1, y_1)$$

$$y_{n+1} = y_n + h f(x_n, y_n)$$



Solve the following differential equation with some initial condition by Euler's method.

$$\frac{dy}{dx} = x + y, y(0) = 1, \text{ hence find } y(0.2), y(0.4), y(0.6).$$

$$\text{Given } f(x, y) = \frac{dy}{dx} = x + y$$

$$y(0) = 1 \Rightarrow y(x_0) = y_0$$

$$\Rightarrow x_0 = 0, y_0 = 1$$

$$x_0 = 0 \quad x_1 = 0.2 \quad x_2 = 0.4 \quad x_3 = 0.6$$

$$y_0 = 1 \quad y_1 = \quad y_2 = \quad y_3 =$$

We have the Euler's formula

$$y_{n+1} = y_n + hf(x_n, y_n)$$

$n=0$

$$y_1 = y_0 + hf(x_0, y_0)$$

$$= 1 + 0.2 \times 1$$

$$= 1.2$$

$$x_0 = 0$$

$$y_0 = 1$$

$$f(x, y) = x + y$$

$$f(x_0, y_0) = x_0 + y_0$$

$$= 0 + 1$$

$$= 1$$

$n=1$

$$y_2 = y_1 + hf(x_1, y_1)$$

$$= 1.2 + 0.2 \times 1.4$$

$$= 1.2 + 0.28$$

$$= 1.48$$

$$f(x_1, y_1) = x_1 + y_1$$

$$= 0.2 + 1.2$$

$$= 1.4$$

$n=2$

$$y_3 = y_2 + hf(x_2, y_2)$$

$$= 1.48 + 0.2 \times 1.88$$

$$= 1.48 + 0.376$$

$$= 1.856$$

$$f(x_2, y_2) = x_2 + y_2$$

$$= 0.4 + 1.48$$

$$= 1.88$$

Modified Euler's Method

$$y_{n+1} = y_n + hf\left[x_n + \frac{h}{2}, y_n + \frac{h}{2}f(x_n, y_n)\right]$$

$n=0$

$$y_1 = y_0 + hf\left[x_0 + \frac{h}{2}, y_0 + \frac{h}{2}f(x_0, y_0)\right]$$

$n=1$

$$y_2 = y_1 + hf\left[x_1 + \frac{h}{2}, y_1 + \frac{h}{2}f(x_1, y_1)\right]$$

etc.

Solve the differential equation with some initial conditions with Modified Euler's method to get the values at $x=0.1$ and $x=0.2$. The differential equation

$$\frac{dy}{dx} = xy, \quad y(0) = 1$$

$$f(x, y) = \frac{dy}{dx} = xy$$

$$h = 0.1 - 0$$

$$= 0.1$$

$$x \quad x_0 = 0 \quad x_1 = 0.1 \quad x_2 = 0.2$$

$$y \quad y_0 = 1 \quad y_1 = \quad y_2 =$$

$$y(0.1) = y_1 = y_0 + hf\left[x_0 + \frac{h}{2}, y_0 + \frac{h}{2}f(x_0, y_0)\right]$$

$$= 1 + 0.1 f\left[0 + \frac{0.1}{2}, 1 + \frac{0.1}{2}f(0, 1)\right] \quad f(0, 1)$$

$$= 1 + 0.1 f\left[0 + \frac{0.1}{2}, 1 + \frac{0.1}{2} \times 0\right] \quad = 0 \times 1 = 0$$

$$= 1 + 0.1 f[0.05, 1] \quad f(0.05, 1)$$

$$= 1 + 0.1 \times 0.05 \quad = 0.05 \times 1$$

$$= 1 + 0.005 = 1.005 \quad = 0.05$$

$$\begin{aligned} y_2(0.2) &= y_1 + h f(x_1 + h/2, y_1 + h/2 f(x_1, y_1)) \\ &= 1.005 + \frac{0.1}{2} f(0.1 + \frac{0.1}{2}, 1.005 + \frac{0.1}{2} f(0.1, 1.005)) \\ &= 1.005 + 0.05 f(0.1 + 0.05, 1.005 + 0.05 \times 0.1005) \\ &= 1.005 + 0.05 f(0.15, 1.010025) \\ &= 1.005 + 0.05 \times 0.15150375 \\ &= 1.005 + 0.0075751875 \quad f(0.1, 1.005) \\ &= 1.0125751875 \quad = 0.1 \times 1.005 \\ & \quad = 0.1005 \end{aligned}$$

4th Order Runge Kutta Method (R-K)
to solve First Order Differential Equation

Algorithm.

First calculate

$$k_1 = hf(x_0, y_0)$$

$$k_2 = hf(x_0 + h/2, y_0 + k_1/2)$$

$$k_3 = hf(x_0 + h/2, y_0 + k_2/2)$$

$$k_4 = h f(x_0 + h/2, y_0 + k_3/2)$$

and compute

$$k = \frac{1}{6} [k_1 + 2k_2 + 2k_3 + k_4]$$

Start from x, y_1 to find y_2

then

$$k_1 = hf(x_1, y_1)$$

$$k_2 = hf(\alpha_1 + h/2, y_1 + k_1/2)$$

$$k_3 = hf(x_2 + h/2, y_2 + k_2/2)$$

$$k_A = hf(x_3 + h/2, y_3 + k_3/2)$$

then the approximate value of y_1 can be found out by $y_1 = y_0 + k$

Find by 4th Order RK Method an approximate value of y for $x=0.2$ in steps of 0.1 if $\frac{dy}{dx} = x+y^2$, $y(0)=1$

if $\frac{dy}{dx} = x + y^2$, $y(0) = 1$

$$\begin{array}{lll} x & x_0 = 0 & x_1 = 0.1 \quad x_2 = 0.2 \quad h = 0.1 - 0 \\ & & & = 0.1 \\ y & y_0 = 1 & y_1 = \quad y_2 = \end{array} \quad f(x,y) = x+y$$

We have the 4th order algorithm a

$$k_1 = hf(x_0, y_0)$$

$$= 0.1 f(0, 1)$$

$$= 0.1 \times 1$$

$$\begin{aligned} f(x_0, y_0) &= f(0, 1) \\ &= 0 + 1^2 \\ &= 1 \end{aligned}$$

$$\begin{aligned} k_2 &= h f(x_0 + h/2, y_0 + k_1/2) \\ &= 0.1 f(0 + \frac{0.1}{2}, 1 + 0.1/2) \\ &= 0.1 f(0.05, 1.05) \\ &= 0.11525 \end{aligned}$$

$$\begin{aligned} f(0.05, 1.05) &= 0.05 + 1.05 \\ &= 1.1525 \end{aligned}$$

$$\begin{aligned}
 k_3 &= hf(x_0 + h/2, y_0 + k_2/2) \\
 &= 0.1 f(0.1 + \frac{0.1}{2}, 1 + 0.11525/2) \\
 &= 0.1 f(0.05, 1.057625) \\
 &= 0.1 \times 1.168570641 \\
 &= 0.116857064
 \end{aligned}$$

$$\begin{aligned}
 k_4 &= hf(x_0 + h/2, y_0 + k_3/2) \\
 &= 0.1 f(0.1 + \frac{0.1}{2}, 1 + 0.116857064/2) \\
 &= 0.1 f(0.05, 1.05843) \\
 &= 0.1 \times 1.17027 \\
 &= 0.117027
 \end{aligned}$$

$$\begin{aligned}
 k &= \frac{1}{6} [k_1 + 2k_2 + 2k_3 + k_4] \\
 &= \frac{1}{6} [0.1 + 2 \times 0.11525 + 2 \times 0.116857064 + 0.117027] \\
 &= 0.11354
 \end{aligned}$$

$$\begin{aligned}
 y_1 &= y_0 + k \\
 &= 1 + 0.11354 \\
 &= 1.11354
 \end{aligned}$$

$$\begin{aligned}
 k_1 &= hf(x_1, y_1) \\
 &= 0.1 f(0.1, 1.11354) \\
 &= 0.1 \times 1.33997 \\
 &= 0.133997
 \end{aligned}$$

$$\begin{aligned}
 k_2 &= hf(x_1 + h/2, y_1 + k_1/2) \\
 &= 0.1 f(0.1 + \frac{0.1}{2}, 1.11354 + 0.133997/2) \\
 &= 0.1 f(0.15, 1.1805386) \\
 &= 0.1 \times 1.54367 \\
 &= 0.154367
 \end{aligned}$$

$$\begin{aligned}
 k_3 &= hf(x_1 + h/2, y_1 + k_2/2) \\
 &= 0.1 f(0.1 + \frac{0.1}{2}, 1.11354 + \frac{0.154367}{2}) \\
 &= 0.1 f(0.15, 1.1907235) \\
 &= 0.1 \times 1.56782 \\
 &= 0.156782
 \end{aligned}$$

$$\begin{aligned}
 k_4 &= hf(x_1 + h/2, y_1 + k_3/2) \\
 &= 0.1 f(0.1 + \frac{0.1}{2}, 1.11354 + \frac{0.156782}{2}) \\
 &= 0.1 f(0.15, 1.1713791) \\
 &= 0.1 \times 1.522129 \\
 &= 0.1522129
 \end{aligned}$$

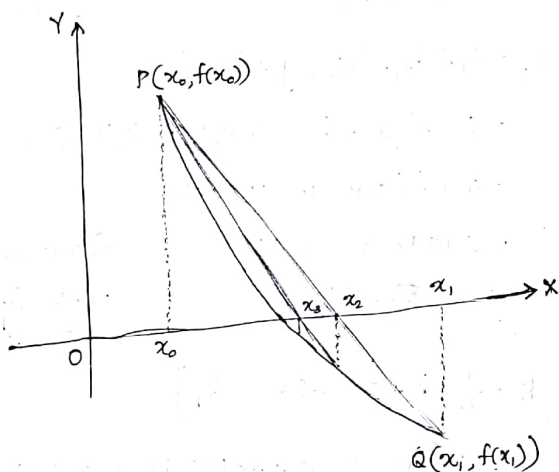
$$\begin{aligned}
 k &= \frac{1}{6} [k_1 + 2k_2 + 2k_3 + k_4] \\
 &= \frac{1}{6} [0.133997 + 2 \times 0.154367 + 2 \times 0.156782 + 0.1522129] \\
 &= 0.137717
 \end{aligned}$$

$$\begin{aligned}
 y_2 &= y_1 + k \\
 &= 1.11354 + 0.137717 \\
 &= 1.25126
 \end{aligned}$$

Assignment
Write 2nd order R-K algorithm with one example.

Regula Falsi Method / Method of False Position

Aim To find the real root of the equation $y = f(x) = 0$.



$$\frac{y - f(x_0)}{x - x_0} = \frac{f(x_1) - f(x_0)}{x_1 - x_0}$$

From the figure it is clear that $y = f(x)$ be a continuous function in the interval (x_0, x_1) such that $f(x_0)$ and $f(x_1)$ are in opposite signs.

This implies that a root lies between x_0 and x_1 .

Equation of the straight line PQ is

$$\frac{y - f(x_0)}{x - x_0} = \frac{f(x_1) - f(x_0)}{x_1 - x_0}$$

$$\Rightarrow y - y_0 = \left[\frac{f(x_1) - f(x_0)}{x_1 - x_0} \right] (x - x_0) \quad \text{--- (1)}$$

From figure it is clear that the line PQ cuts on x-axis at a point $(x_2, 0)$.

$$\therefore \text{①} \Rightarrow 0 - f(x_0) = \left[\frac{f(x_1) - f(x_0)}{x_1 - x_0} \right] (x_2 - x_0)$$

$$x_2 = x_0 + \frac{f(x_1) - f(x_0)}{f(x_1) - f(x_0)} \cdot \frac{f(x_0)}{1}$$

$$x_2 = x_0 + \frac{x_1 - x_0}{f(x_1) - f(x_0)} f(x_0)$$

$$x_2 = x_0 + \frac{x_1 f(x_0) - x_0 f(x_1)}{f(x_1) - f(x_0)}$$

This procedure is repeated until we get the appropriate root.

Find the root of the equation

$$2x - \log_{10} x = 7 \text{ which lies between } 3.5$$

and 4 corrected to five decimal places using Regula Falsi Method in two iterations.

$$2x - \log_{10} x = 7$$

$$2x - \log_{10} x - 7 = 0$$

$$f(x) = 0$$

$$\text{where } f(x) = 2x - \log_{10} x - 7$$

$$\text{Also given } x_0 = 3.5$$

$$x_1 = 4$$

$$f(x_0) = f(3.5) = 2 \times 3.5 - \log_{10} 3.5 - 7$$

$$= -0.54407$$

$$f(x_1) = f(4) = 2 \times 4 - \log_{10} 4 - 7$$

$$= 0.39794$$

clearly $f(3.5)$ and $f(4)$ are opposite in sign therefore a root lies between 3.5 and 4. Hence we use the iterative formula

$$x_2 = x_0 + \frac{x_0 f(x_1) - x_1 f(x_0)}{f(x_1) - f(x_0)}$$

$$= 3.5 + \frac{3.5 \times 0.39794 - 4 \times (-0.54407)}{0.39794 - (-0.54407)}$$

$$= 3.78878$$

$$f(3.78878) = 2 \times 3.78878 - \log_{10} 3.78878 - 7$$

$$= -0.00104$$

$f(1)$ and $f(2)$ are on opposite signs

$$x_0 = 3.78878$$

$$x_1 = 4$$

$$x_3 = \frac{3.78878 \times 0.39794 - 4 \times (-0.00104)}{0.39794 - (-0.00104)}$$

$$= 3.78934$$

Find by the method of Regula Falsi a root of the equation $x^3 + x^2 - 3x - 3 = 0$ lies between 1 and 2 upto three iterations.

$$f(x) = x^3 + x^2 - 3x - 3 = 0$$

$$f(x) = x^3 + x^2 - 3x - 3$$

$$x_0 = 1$$

$$x_1 = 2$$

$$f(x_0) = f(1) = 1^3 + 1^2 - 3(1) - 3$$

$$= 1 + 1 - 3 - 3$$

$$= -4 < 0$$

$$f(x_1) = f(2) = 2^3 + 2^2 - 3(2) - 3$$

$$= 8 + 4 - 6 - 3$$

$$= 3 > 0$$

$$x_2 = \frac{x_0 f(x_1) - x_1 f(x_0)}{f(x_1) - f(x_0)}$$

$$= \frac{1 \times 3 - 2 \times (-4)}{3 - (-4)}$$

$$= \frac{3 + 8}{7} = \frac{11}{7} = 1.571$$

$$f(1.571) = (1.571)^3 + (1.571)^2 - 3(1.571) - 3$$

$$= -1.368 < 0$$

∴ Root lies between 1.571 and 2

$$x_0 = 1.571$$

$$x_1 = 2$$

$$\therefore x_3 = \frac{1.571 \times 3 - 2 \times -1.368}{3 - (-1.368)}$$

$$= 1.705$$

$$f(1.705) = (1.705)^3 + (1.705)^2 - 3(1.705) - 3$$

$$= -0.247 < 0$$

Root lies between 1.705 and 2.

$$x_0 = 1.705$$

$$x_1 = 2$$

$$x_4 = \frac{1.705 \times 3 - 2 \times -0.247}{3 - (-0.247)}$$

$$= 1.727$$

Newton Raphson Method

Aim

To find the root of the equation $f(x) = 0$

Procedure

Let x_0 be a point near the root of the equation. Then equation of the

tangent at $P(x_0, f(x_0))$ is

$$y = f(x_0) + f'(x_0)(x - x_0) \quad \text{--- ①}$$

Let the tangent at $(x_0, f(x_0))$ cuts x axis at $(x_1, 0)$

then ① $\Rightarrow$

$$0 - f(x_0) = f'(x_0)(x_1 - x_0)$$

$$\Rightarrow x_1 = x_0 - \frac{f(x_0)}{f'(x_0)}$$

Similarly

$$x_2 = x_1 - \frac{f(x_1)}{f'(x_1)}$$

and hence

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

Find the real root of the equation $3x = \cos x + 1$ by using Newton Raphson method.

$$3x = \cos x + 1$$

$$\Rightarrow \cos x - 3x + 1 = 0 \quad \text{or}$$

$$f(x) = \cos x - 3x + 1 \quad \text{or} \quad 3x - \cos x - 1 = 0$$

$$3x - \cos x - 1$$

$$\text{Clearly, } f(0) = -\cos 0 + 3(0) + 1$$

$$= -1 + 0 + 1$$

$$= 0$$

$$f(1) = -\cos 1 + 3(1) + 1$$

$$= 1.4597 > 0$$

∴ A root lies between 0 and 1

$$f(x) = 3x - \cos x - 1$$

$$f(x_n) = 3x_n - \cos(x_n) - 1$$

$$f'(x_n) = 3 + \sin x_n$$

∴ By Newton Raphson Method

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

$$= x_n - \frac{3x_n - \cos x_n - 1}{3 + \sin x_n}$$

$$x_{n+1} = \frac{x_n \sin x_n + \cos x_n + 1}{3 + \sin x_n} \quad \text{--- (2)}$$

Put $n=0$ in (2)

$$x_1 = \frac{x_0 \sin x_0 + \cos x_0 + 1}{3 + \sin x_0}$$

$$= \frac{0.6 \sin(0.6) + \cos(0.6) + 1}{3 + \sin(0.6)}$$

~~0.6071~~

$$= 0.6071$$

Then

$$x_2 = \frac{x_1 \sin x_1 + \cos x_1 + 1}{3 + \sin x_1}$$

$$= \frac{0.6071 \times \sin(0.6071) + \cos(0.6071) + 1}{3 + \sin(0.6071)}$$

$$= 0.6071$$

$$x_1 = x_2$$

∴ 0.6071 is a root of the equation

$$f(x) = 0.$$

Assume $x_0 = 0.6$

Since root lies between 0 and 1

Find a positive real number root of $x^3 + 2x^2 + 10x - 20 = 0$ using Newton Raphson Method.

$$f(x) = x^3 + 2x^2 + 10x - 20 = 0$$

$$f(0) = (0)^3 + 2(0)^2 + 10(0) - 20$$

$$= -20$$

$$f(1.3) = (1.3)^3 + 2(1.3)^2 + 10(1.3) - 20$$

$$= -1.113$$

$$f(1.4) = (1.4)^3 + 2(1.4)^2 + 10(1.4) - 20$$

$$= 0.664$$

$$f(x_n) = x_n^3 + 2x_n^2 + 10x_n - 20$$

$$f'(x_n) = 3x_n^2 + 4x_n + 10$$

$$\therefore x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

$$= x_n - \frac{x_n^3 + 2x_n^2 + 10x_n - 20}{3x_n^2 + 4x_n + 10}$$

$$= \frac{3x_n^3 + 4x_n^2 + 10x_n - (x_n^3 + 2x_n^2 + 10x_n - 20)}{3x_n^2 + 4x_n + 10}$$

$$= \frac{2x_n^3 + 2x_n^2 + 20}{3x_n^2 + 4x_n + 10}$$

$$= \frac{2(x_n^3 + x_n^2 + 10)}{3x_n^2 + 4x_n + 10}$$

Put $n = 0, 1, 2,$

Assume $x_0 = 1.3$

$$n=0 \quad x_1 = \frac{2(x_0^3 + x_0^2 + 10)}{3x_0^2 + 4x_0 + 10}$$

$$= \frac{2((1.3)^3 + (1.3)^2 + 10)}{3(1.3)^2 + 4(1.3) + 10}$$

$$= \frac{27.774}{20.27}$$

$$= 1.3702$$

$n=1$

$$x_2 = \frac{2(x_1^3 + x_1^2 + 10)}{3x_1^2 + 4x_1 + 10}$$

$$= \frac{2((1.3702)^3 + (1.3702)^2 + 10)}{3(1.3702)^2 + 4(1.3702) + 10}$$

$$= \frac{28.8999}{21.1131}$$

$$= 1.3688$$

$n=2$

$$x_3 = \frac{2(x_2^3 + x_2^2 + 10)}{3x_2^2 + 4x_2 + 10}$$

$$= \frac{2((1.3688)^3 + (1.3688)^2 + 10)}{3(1.3688)^2 + 4(1.3688) + 10}$$

$$= \frac{28.8764}{21.096}$$

$$= 1.3688$$

$$x_4 = 1.3688$$

$$x_5 = 1.3688$$

$\therefore 1.3688$ is the root of the equation
 $f(x)=0$.

Apply Gauss Seidel method to solve

$$20x + y - 2z = 17$$

$$3x + 20y - z = -18$$

$$2x - 3y + 20z = 25$$

$$x = \frac{1}{20} [17 - y + 2z]$$

$$y = \frac{1}{20} [-18 - 3x + z]$$

$$z = \frac{1}{20} [25 - 2x + 3y]$$

~~Choose~~

Choose the initial values as $y_0 = 0, z_0 = 0$

$$x_1 = \frac{1}{20} [17 - 0 + 0]$$

$$= 0.85$$

Put $x_1 = 0.85$ and $z_0 = 0$ in ①

$$y_1 = \frac{1}{20} [-18 - 3(0.85) + 0]$$

$$= -1.0275$$

Put $x = x_1 = 0.85$, and $y = y_1 = -1.0275$ in ③

$$z_1 = \frac{1}{20} [25 - 2(0.85) + 3(-1.0275)]$$

$$= 1.0109$$

Second Iteration

Start with $x_1 = 0.85, y_1 = -1.0275, z_1 = 1.0109$
then find x_2, y_2, z_2 .

$$x_2 = \frac{1}{20} [17 - (-1.0275) + 2(1.0109)]$$

$$= 1.0025$$

$$y_2 = \frac{1}{20} [-18 - 3(1.0025) + 1.0109]$$

$$= -0.9998$$

$$z_2 = \frac{1}{20} [25 - 2(1.0025) + 3(-0.9998)]$$

$$= 0.9999$$

$$x_1 = \frac{1}{20} [17 - 2(0.85) + 3(-)]$$

Third iteration

Start with $x_2 = 1.0025, y_2 = -0.9998$

$$x_3 =$$

$$= 1.0000$$

$$y_3 =$$

$$= 1.0000$$

$$z_3 =$$

$$= 1.0000$$

Hence solution of the Gauss Seidel method
is $x = 1, y = -1$ and $z = 1$

$$2x_1 - x_2 + 2x_3 = 3$$

$$x_1 + 3x_2 + 3x_3 = -1$$

$$x_1 + 2x_2 + 5x_3 = 1$$

- With initial values

$$x_1 = 0.3, x_2 = -0.8$$

$$x_3 = 0.3$$